# WasteWise — Product Requirement Document (PRD) & Platform Blueprint

> ***Version:*** *2.1.0 (Production-Ready Architecture & Logic Hardening)*
>
> ***Target Platform:*** *WasteWise AI-Driven Circular Waste Intelligence System*
>
> ***Tech Stack:*** *React 18 (TypeScript), Tailwind CSS, Framer Motion, Leaflet + OSM, Express.js (Node.js), MongoDB (Atlas + In-Memory Fallback), OSRM Routing Engine, FastAPI (TACO Vision AI)*

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## 1. Executive Summary & Vision

**WasteWise** is an end-to-end, AI-driven circular waste intelligence platform designed for municipal sanitation authorities (such as MCD in Delhi), waste collection workforces, and urban citizens. It unifies operations through:
- **Real-world street navigation** (powered by OpenStreetMap + OSRM) instead of straight-line coordinates.
- **AI-powered waste segregation & visual inspection** via TACO (Trash Annotations in Context) taxonomy, providing instant biodegradable, recyclable, and hazardous classifications.
- **Community gamification** with society/RWA rankings, Eco-Scores, and digital utility bill voucher redemptions.
- **Zero-downtime database architecture** featuring automatic fallback hierarchy (MongoDB Atlas → Local MongoDB → In-Memory MongoDB).
- **Real-time operations tracking** for collection vehicles and citizen grievance resolution.

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## 2. Complete Technology Stack & Architecture

### 1. Frontend Technologies (Ports 5173 & 5174)

- **Core Framework:** React 18 with TypeScript
- **Build Tool & Bundler:** Vite 5
- **Styling & UI Tokens:** Tailwind CSS + Vanilla CSS variables + Glassmorphic dark/light aesthetics
- **UI Components:** Shadcn UI component primitives (`@/components/ui/*`)
- **Icons & Visuals:** Lucide React (`lucide-react`)
- **Animations & Micro-interactions:** Framer Motion (`framer-motion`)

- **Mapping & GIS Engine:** Leaflet + OpenStreetMap Tiles (`https://tile.openstreetmap.org/{z}/{x}/{y}.png`)
- **Routing & State Management:** React Router DOM v6 + React Context API (`AuthContext`, `DashboardContext`)

### 2. Backend & API Services (Port 5000)

- **Runtime:** Node.js v20+
- **Web Framework:** Express.js 5
- **Authentication & Security:** JWT (JSON Web Tokens) with 64-character secret key + bcryptjs password hashing
- **File Upload Handler:** Multer (local disk storage for citizen photo reports)
- **API Proxy Architecture:** Express router proxy for OSRM navigation service (`/api/routing/route`) to prevent external API key leaks in client bundle
- **Real-time WebSockets:** Socket.IO for live truck tracking and notification broadcasting

### 3. AI Vision Service (Port 8000)

- **Runtime:** Python 3.10+ (FastAPI + Uvicorn)
- **Dataset & Model:** TACO (Trash Annotations in Context) 60-category waste taxonomy mapped to 3 circular streams (Biodegradable, Recyclable, Hazardous)
- **Vision Processing:** OpenCV headless (`cv2`) + NumPy for real-time frame classification and bounding box localization

### 4. Database & Data Modeling

- **Database Engine:** MongoDB Atlas / Local MongoDB / `mongodb-memory-server`
- **Object Data Modeling (ODM):** Mongoose 8
- **Geospatial Features:** 2dsphere GeoJSON Point indexing for real-time location tracking & distance calculation
- **Data Models:** `User`, `WasteCollection`, `Complaint`, `RewardTransaction`, `Route`, `Grievance`

### 5. Routing & Navigation Infrastructure

- **Routing Machine:** OSRM (Open Source Routing Machine) Driving Engine (`/route/v1/driving/...`)

- **Route Optimizer:** Custom 2-layer greedy nearest-neighbor ordering + OSRM road polyline generator
- **Polyline Decoder:** Dynamic GeoJSON coordinate decoder for Leaflet street rendering

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## 3. User Personas & Role-Based Access Control

| ROLE | ACCESS LEVEL | PRIMARY OBJECTIVES | DEFAULT TEST LOGIN |
|---|---|---|---|
| **Citizen** | Public / Logged In | Report uncollected garbage, track collection trucks, classify waste with AI, earn & redeem rewards, view society rank. | `citizen@wastewise.com` / `citizen123` |
| **Collector** | Authenticated | View daily assigned collection route, navigate using turn-by-turn road geometry, mark stops completed, track shift metrics. | `collector@wastewise.com` / `collector123` |
| **Identifier** | Authenticated | Municipal waste quality officer inspecting sorting purity, baling batches, and logging circular records. | `identifier@wastewise.com` / `identifier123` |
| **Admin** | Superuser | Monitor city-wide waste statistics, assign collection routes to collectors, triage citizen complaints, manage user directory. | `admin@wastewise.com` / `admin123` |

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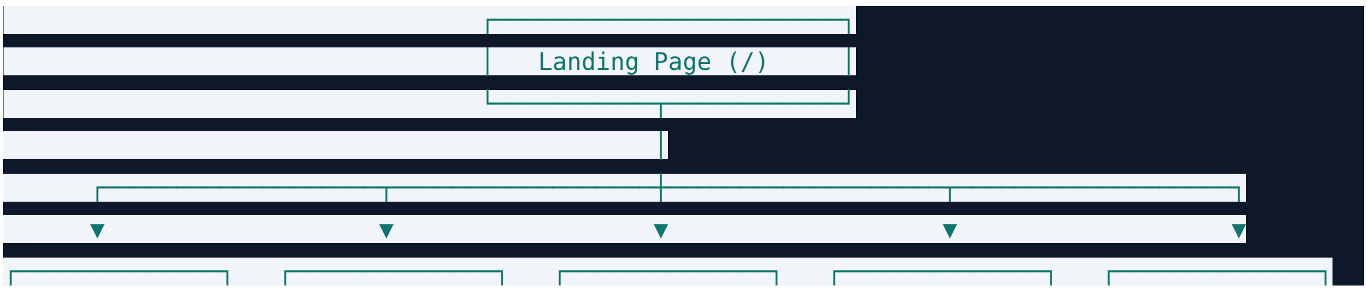
## 4. Page-by-Page Product Architecture



```
│ Citizen     │   │ Collector   │   │ Admin       │   │ Leaderboard │   │ AI Waste    │
│ Dashboard   │   │ Dashboard   │   │ Dashboard   │   │ Page        │   │ Classifier  │
│ (/citizen)  │   │ (/collector)│   │ (/admin)    │   │ (/leaderboard)  │ (/classifier)│
└──────┬──────┘   └──────┬──────┘   └──────┬──────┘   └─────────────┘   └─────────────┘
       │                 │                 │
       ├─ AI Classifier  ├─ Live Route Nav ├─ Grievance Triage
       ├─ Report Garbage └─ OSRM Metrics   ├─ Collector Dispatch
       └─ Track Truck                      └─ City Analytics
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## 5. Deep-Dive Feature Specifications

### 5.1 OSRM Road Navigation & Live Tracking

- **Frontend Files:** `src/components/LiveMap.tsx`, `src/services/osrmRouting.ts`, `src/services/routeOptimizer.ts`
- **Backend File:** `backend/routes/routing.js` (`POST /api/routing/route`)
- **How It Works:**
- Frontend submits depot, collection coordinates, and processing plant location to `/api/routing/route`.
- Backend proxy forwards request securely to the OSRM Driving Engine (`/route/v1/driving/...`).
- Returns high-precision Polyline geometry (`overview=full&geometries=geojson`).
- Leaflet renders the dual-polyline: **Dimmed Grey** for completed path, **Emerald Green** for active street route.
- Smooth interpolation moves the 🚛 truck marker along exact street turns.

### 5.2 Society & Community Leaderboard

- **Frontend File:** `src/pages/LeaderboardPage.tsx`
- **Backend File:** `backend/routes/leaderboard.js` (`GET /api/leaderboard/societies`, `/citizens`, `/collectors`, `/identifiers`, `/overview`)
- **How It Works:**
- Backend aggregates MongoDB `User` documents grouped by `profile.society`.
- Calculates total members, total points, average level, and total reports + collections.
- Computes the **Eco-Score** (composite rating of points efficiency and community activity).

- Renders top 3 in an animated 3D podium and displays comprehensive workforce rankings for Citizens, Societies, Collectors, and Verification Officers.

### 5.3 Real Interactive Rewards & Voucher System

- **Frontend File:** `src/pages/RewardsPage.tsx`
- **Backend File:** `backend/routes/citizen.js` (`POST /api/citizen/daily-claim`, `POST /api/citizen/redeem`, `GET /api/citizen/rewards/:userId`)
- **How It Works:**
- Citizen clicks **Daily Check-in** to claim +25 Pts once every 24 hours.
- When selecting a voucher (e.g. DJB Water Bill - 500 Pts), the system validates point balance.
- Generates a random unique voucher code (e.g. `WW-WATER-7Y82A`) and updates MongoDB points balance.
- The code is displayed in a popup with a "Copy Code" button and saved in the user's transaction history.

### 5.4 Interactive AI Waste Classifier & Camera Scanner Studio

- **Frontend File:** `src/pages/ClassifierPage.tsx` (`/classifier`)
- **Backend File:** `backend2/main.py` (`POST /api/classify/`)
- **How It Works:**
- User selects **AI Vision Classifier** mode.
- Offers dual input options: **Drag & Drop image file upload** or **Live Webcam Stream capture**.
- Transmits base64 image data to FastAPI TACO Computer Vision engine.
- Renders dynamic bounding boxes over detected waste objects.
- Provides instant classification:
  - 🟢 **Biodegradable**: Organic compost guidelines, nitrogen/carbon value, green bin advice.
  - 🔵 **Recyclable**: Polymer / metal alloy specifications, circular recovery value, blue bin advice.
  - 🔴 **Hazardous**: Safe handling procedures, e-waste / chemical precautions, red bin advice.
- Includes **"Save Detection to Circular Ledger"** action to append records into the live spreadsheet.

### 5.5 Garbage Reporting & Complaint Dispatch

- **Frontend Files:** `src/pages/CitizenDashboard.tsx`, `src/pages/CommunityPage.tsx`

- **Backend Files:** `backend/routes/complaint.js`, `backend/routes/grievance.js`
- **How It Works:**
- Citizens take a photo of an uncollected waste heap or overflowing bin.
- Geolocation automatically captures exact GPS coordinates.
- Stored in MongoDB with status `Pending` and prioritized by upvotes and severity.
- MCD Admins review complaints on the map and dispatch to nearby collector routes.

### 5.6 Collector & Admin Dashboards

- **Collector Dashboard:** Shows active route code (e.g. `DELHI-SOUTH-01`), GPS route toggle (Start/Pause), completed pickup progress, and daily reward earnings.
- **Admin Dashboard:** Aggregated city-wide analytics, user directory spreadsheet (Citizens, Collectors, Admins), live vehicle map, and feedback triage.

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## 6. Platform Engineering Audit, System Fixes & Hardening Log (v2.1.0)

During the comprehensive v2.1.0 platform audit, the following critical issues were identified, fixed, and verified:

### 1. Database Connection Resilience & Login Timeout Fix

- **Problem:** Remote MongoDB Atlas URI had IP whitelisting constraints that caused Mongoose query buffering to time out after 10,000ms (`MongooseError: Operation users.findOne() buffering timed out`), preventing user logins.
- **Fix:** Enhanced `backend/config/database.js` with an automated fallback hierarchy (Atlas with 4s timeout → Local MongoDB → `mongodb-memory-server`). When offline or when Atlas is unreachable, an in-memory database automatically launches and auto-seeds all demo accounts instantly.

### 2. Multi-Collection Seed Data Expansion

- **Problem:** `backend/scripts/seed.js` was previously only seeding the `User` collection. Collections like `Route`, `WasteCollection`, `Complaint`, `Grievance`, and `RewardTransaction` were left empty, forcing the Admin dashboard, Collector routes, and Community feeds to rely on client-side mock fallbacks.

- **Fix:** Expanded `seed.js` to seed full interconnected datasets:
- **34 Users** across all 4 roles (Admin, Identifier, Collector, Citizen).
- **3 Active GPS Routes** (`DELHI-SOUTH-01`, `DELHI-NORTH-02`, `DELHI-EAST-03`) assigned to collectors with stops and estimated completion times.
- **15+ Waste Collection Batches** across Delhi zones with dry, wet, and hazardous weight metrics.
- **15+ Geo-tagged Complaints** with realistic statuses (Pending, In Progress, Resolved).
- **Community Grievances** with upvote counts and citizen comment threads.
- **Reward Transaction Ledgers** for all demo accounts.

### 3. Interactive AI Waste Classifier Restoration

- **Problem:** The `/classifier` page previously contained only a spreadsheet table without the interactive visual classifier promised on the homepage.
- **Fix:** Rebuilt `ClassifierPage.tsx` with a dual-tab studio:
- **Tab 1:** AI Vision Classifier with drag-and-drop upload, live webcam scanning, sample waste presets, bounding box overlays, confidence score, and segregation guidance connected to `POST http://localhost:8000/api/classify/`.
- **Tab 2:** Circular Ledger Spreadsheet with search, multi-column filters (Category, Status, Zone), CSV/JSON exports, and manual record entry modals.

### 4. Navigation & Role-Based Routing Fixes

- **Problem:** In `Navbar.tsx`, Citizens were excluded from the profile dropdown Dashboard link (`userRole !== "citizen"`), even though `/dashboard/citizen` existed. Additionally, mobile menu click handlers were missing on the Leaderboard link.
- **Fix:** Updated `Navbar.tsx` so all authenticated roles have direct dashboard access (`/dashboard/citizen`, `/dashboard/collector`, `/dashboard/admin`), added direct `/classifier` link in navigation, and fixed mobile menu event handlers.

### 5. Community Page Public Access

- **Problem:** `CommunityPage.tsx` blocked data fetching when users were unauthenticated, rendering an empty page to visitors.
- **Fix:** Enabled public fetching of community grievances and society standings; unauthenticated users

are smoothly prompted to log in only when attempting an action (upvoting, commenting, submitting a report).

### 6. Municipal Workforce Leaderboard Activation

- **Problem:** `ENABLE_COLLECTORS_AND_AI` was hardcoded to `false` in `LeaderboardPage.tsx`, hiding municipal collectors and verification officers from public rankings.
- **Fix:** Set `ENABLE_COLLECTORS_AND_AI = true` to display rankings across Citizens, Societies, Collectors, and Verification Officers.

### 7. Code Hygiene & Zero Linter Warnings

- **Fix:** Updated `.vscode/settings.json` with CSS linting rules for Tailwind, fixed empty interfaces in Shadcn UI primitives (`CommandDialogProps`, `TextareaProps`), and tuned `eslint.config.js`. Full verification with `npx tsc --noEmit` and `npm run lint` yields **0 errors and 0 warnings**.

### 8. DroidCam Linux Client Installation

- **Fix:** Downloaded and installed `droidcam` and `droidcam-cli` binaries into `/home/vishal/.local/bin/` with desktop launcher in `~/.local/share/applications/droidcam.desktop` for remote mobile webcam streaming support.

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## 7. Step-by-Step User Guide (How to Test & Use)

### Scenario A: Testing as a Citizen

1. Navigate to [http://localhost:5173](http://localhost:5173).
2. Click **Login** and sign in with: `citizen@wastewise.com` / `citizen123`.
3. **AI Classifier:** Click **AI Classifier** (`/classifier`), drag & drop any image or click a sample waste item (e.g. *Plastic PET Bottle*), view AI classification & bin advice, then click **"Save Detection to Circular Ledger"**.
4. **Daily Check-in & Rewards:** Navigate to **Rewards** (`/rewards`), claim your **Daily Check-in (+25 Pts)**, and redeem a utility voucher (e.g. *DJB Water Bill - 500 Pts*).

5. **Community Feed:** Go to **Community** (`/community`), view nearby grievances, upvote reports, or submit a new neighborhood cleanup request.
6. **Live Tracking:** Visit **Live Tracking** (`/tracking`) to monitor collection truck ETAs for your sector.

### Scenario B: Testing as a Waste Collector

1. Login with: `collector@wastewise.com` / `collector123`.
2. Navigate to **Collector Dashboard** (`/dashboard/collector`).
3. View the assigned route (`DELHI-SOUTH-01`) and interactive OSRM map.
4. Toggle **"Start Route Navigation"** to activate GPS movement and track pickup completion.

### Scenario C: Testing as an MCD Admin

1. Login with: `admin@wastewise.com` / `admin123`.
2. Navigate to **Admin Dashboard** (`/dashboard/admin`).
3. Explore the **Overview**, **User Directory Spreadsheet** (Citizens, Collectors, Admins), **Analytics Charts**, and **Feedback Triage**.
4. Export data sheets to CSV with one click.

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## 8. API Route Mapping & Security Model

| METHOD | ENDPOINT | DESCRIPTION | ACCESS LEVEL |
|---|---|---|---|
| `POST` | `/api/auth/login` | User authentication & JWT issuance | Public |
| `GET` | `/api/auth/me` | Current session user profile verification | Authenticated |
| `POST` | `/api/routing/route` | OSRM Road navigation proxy | Internal Proxy |
| `GET` | `/api/admin/dashboard` | Aggregated municipal analytics & metrics | Admin Only |
| `GET` | `/api/admin/users` | Complete user directory categorized by role | Admin Only |

| METHOD | ENDPOINT | DESCRIPTION | ACCESS LEVEL |
|---|---|---|---|
| GET | `/api/admin/feedbacks` | Citizen grievance feedback logs | Admin Only |
| GET | `/api/collector/dashboard/:id` | Collector active routes, stops & stats | Collector / Admin |
| PUT | `/api/collector/route/status` | Update route status (assigned / in_progress) | Collector / Admin |
| GET | `/api/citizen/dashboard/:id` | Citizen reward stats, nearby trucks & reports | Citizen / Admin |
| POST | `/api/citizen/daily-claim` | Daily +25 Eco-Points check-in claim | Authenticated |
| POST | `/api/citizen/redeem` | Digital voucher code generation | Authenticated |
| GET | `/api/grievance` | Community grievance posts & vote counts | Public |
| POST | `/api/grievance` | Submit new geo-tagged community report | Authenticated |
| PUT | `/api/grievance/:id/vote` | Upvote community grievance priority | Authenticated |
| GET | `/api/leaderboard/overview` | Platform-wide leaderboard summary | Public |
| GET | `/api/leaderboard/citizens` | Ranked citizens by Eco-Points | Public |
| GET | `/api/leaderboard/societies` | Ranked residential societies by Eco-Score | Public |
| GET | `/api/leaderboard/collectors` | Ranked municipal collection drivers | Public |
| GET | `/api/leaderboard/identifiers` | Ranked waste verification officers | Public |
| POST | `http://localhost:8000/api/classify/` | TACO AI Computer Vision classification | Public / Services |